Compact Plasma-Assisted Space-Charge Neutralizer for High-Current Cyclotron Axial Injection: Physics Models, Simulation Workflows, and Performance Analysis

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Abstract

High-current (10 mA), low-energy (30 keV) proton beams experience severe uncompensated space-charge forces during transport through low-energy beam transport (LEBT) lines prior to entering a cyclotron spiral inflector. This report consolidates the complete physics theoretical models, atomic collision kinematics, numerical particle-in-cell (PIC) simulation methodologies, diagnostics algorithms, and downstream transport optics for a compact gas-ionized plasma neutralizer cell. We systematically compare Hydrogen (H_2) and Krypton (Kr) neutralizer gases, accounting for laboratory-to-center-of-mass collision kinematics ($\sigma_{\mathrm{Kr}}/\sigma_{\mathrm{H2}} \approx 5.56$ at 30 keV). Four numerical simulation paradigms are benchmarked: vacuum reference transport, static analytical seeded compensation, dynamic Python callback source models, and fully self-consistent C++ Monte Carlo Collision (MCC) PICMI extensions. We demonstrate that operating a Krypton cell at 1.0×10^{-6} Torr achieves space-charge perveance reduction equivalent to 1.0×10^{-5} Torr of H_2 ($\eta_{\mathrm{net}} \geq 90\%$), reducing transverse envelope expansion and raising inflector entrance transmission from $< 35\%$ (uncompensated) to $> 98\%$. Furthermore, we evaluate the impact of RF bunching on peak-bunch perveance degradation ($K_{\mathrm{eff,peak}}/K_{0,\mathrm{peak}} \approx 1 - \eta_{\mathrm{avg}}/B_f$), providing practical design guidelines and publication standards for compact cyclotron injection lines.

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1 Introduction and System Architecture

1.1 Physics Context

High-intensity compact cyclotrons for medical isotope production, hadron therapy, and industrial particle applications require multi-milliampere ion beam currents injected at relatively low kinetic energies ($E_k = 30\,\text{keV}$ for protons, $\beta = v/c \approx 0.00799$). In uncompensated Low-Energy Beam Transport (LEBT) lines, the strong repulsive self-electric field generates severe beam perveance:

$$K_0 = \frac{qI_{\text{avg}}}{2\pi\varepsilon_0 m_p v_{\text{beam}}^3 \gamma^3} \approx 6.43 \times 10^{-4} \quad (\text{for } 30\,\text{keV}, 10\,\text{mA } p^+) \tag{1}$$

Without space-charge mitigation, this high perveance induces rapid transverse beam blowup, causing heavy particle loss on apertures, emittance growth, and poor acceptance matching into the small aperture ($r_{\text{aperture}} \approx 5.0\,\text{mm}$) of the downstream spiral inflector.

1.2 Baseline Axial Injection Beamline Layout

To mitigate space-charge divergence prior to the primary focusing optics, the baseline compact cyclotron axial injection line is structured as:

$$\text{buncher} \longrightarrow \text{plasma neutralizer} \longrightarrow \text{solenoid} \longrightarrow \text{quadrupole Q1} \longrightarrow \text{quadrupole Q2} \longrightarrow \text{spiral inflector} \tag{2}$$

Baseline Layout Architecture

$$[\text{Buncher}] \xrightarrow{\text{bunched beam}} [\text{Plasma Neutralizer Cell}] \xrightarrow{\text{neutralized}} [\text{Solenoid}] \rightarrow [\text{Quad Doublet Q1/Q2}] \rightarrow [\text{Inflector}]$$

Figure 1: Baseline layout of the compact cyclotron axial injection line. The neutralizer cell is located strictly **upstream** of the primary solenoid lens.

Critical Geometric Constraint: The plasma neutralizer cell is located strictly **upstream** of the main solenoid. Placing the neutralizer cell downstream of the solenoid or near the inflector entrance is treated only as an explicit alternative-placement study.

2 Fundamental Physics Models and Collision Kinematics

2.1 Proton-Impact Ionization Dynamics

When a 30 keV proton beam passes through a neutral background gas cell (H_2 or Kr), primary electron-ion pairs are generated via impact ionization:

$$p^+ + \text{Gas} \longrightarrow p^+ + \text{Gas}^+ + e^- \tag{3}$$

Secondary electrons are trapped within the attractive space-charge potential well of the beam, while secondary ions are repelled radially.

2.2 Ionization Time Constant and Neutralization Buildup

The characteristic space-charge neutralization time constant τ is defined as:

$$\tau = \frac{1}{n_{\text{gas}} \sigma_{\text{ion}} v_{\text{beam}}} \tag{4}$$

where $n_{\text{gas}} = p/(k_B T)$ is the neutral gas number density at pressure p and temperature $T = 300\,\text{K}$, σ_{ion} is the total proton-impact ionization cross section, and $v_{\text{beam}} = \beta c = 2.396 \times 10^6\,\text{m/s}$.

The time-dependent buildup of space-charge neutralization follows:

$$\eta(t) = \eta_{\text{ss}} \left(1 - e^{-t/\tau}\right) \tag{5}$$

where η_{ss} is the steady-state equilibrium compensation level ($\approx 0.90 - 0.95$).

2.3 Laboratory vs Center-of-Mass Frame Cross Sections

Cross-section data are parameterized by center-of-mass energy E_{cm}:

$$E_{\text{cm}} = E_{\text{lab}} \cdot \left(\frac{m_{\text{target}}}{m_{\text{projectile}} + m_{\text{target}}}\right) \tag{6}$$

For a $30\,\text{keV}$ ($E_{\text{lab}} = 30,000\,\text{eV}$) proton projectile:

- **H_2 Target** ($m_{\text{target}} \approx 2m_p$): $E_{\text{cm}} = 30,000 \cdot (2/3) = 20,000\,\text{eV} = 20.00\,\text{keV}$.
- **Kr Target** ($m_{\text{target}} \approx 83.8m_p$): $E_{\text{cm}} = 30,000 \cdot (83.8/84.8) \approx 29,646\,\text{eV} = 29.65\,\text{keV}$.

Table 1: Proton-Impact Ionization Cross Sections at 30 keV Laboratory Energy

Target Gas	E_{lab} [keV]	E_{cm} [keV]	σ_{ion} [m^2]	σ_{ion} [Å^2]	Relative Ratio ($\sigma_{\text{Kr}}/\sigma_{\text{H2}}$)
Hydrogen (H_2)	30.0	20.00	1.6135×10^{-20}	1.61	1.00×
Krypton (Kr)	30.0	29.65	8.9648×10^{-20}	8.96	5.56×

Key Takeaway: Because $\sigma_{\text{Kr}} \approx 5.56\,\sigma_{\text{H2}}$, a Krypton gas cell operates at a 5-to-10-fold lower neutral gas pressure (1.0×10^{-6} Torr vs 1.0×10^{-5} Torr H_2) while providing identical plasma generation rates, thereby minimizing emittance growth from elastic gas scattering.

2.4 RF-Bunched Beam Space-Charge Compensation

Because the RF buncher is positioned upstream of the neutralizer, the beam enters the plasma cell already bunched. For an RF frequency f_{RF} (50 MHz) and phase width $\Delta\phi$, the bunching factor is:

$$B_f = \frac{I_{\text{peak}}}{I_{\text{avg}}} = \frac{360^\circ}{\Delta\phi} \tag{7}$$

While plasma electrons form a quasi-steady background providing average neutralization $\eta_{\text{avg}} = N_e/N_p$, the peak-bunch instantaneous charge density is enhanced by B_f. The effective peak perveance reduction ratio is:

$$\frac{K_{\text{eff,peak}}}{K_{0,\text{peak}}} \approx 1 - \frac{\eta_{\text{avg}}}{B_f} \tag{8}$$

For example, at $B_f = 5.0$ and $\eta_{\text{avg}} = 0.90$, $K_{\text{eff,peak}}/K_{0,\text{peak}} \approx 1 - 0.90/5.0 = 0.82$. Thus, while average perveance is reduced by 90%, peak-bunch perveance is reduced by only 18%.

3 Literature Review and Related Work

Space-charge neutralization in low-energy ion transport has been studied across multiple accelerator disciplines:

1. **Cyclotron Axial Injection Lines**: Early work by Jongen et al. [1] and Kazarinov et al. [2, 3] highlighted space-charge limitations in high-current heavy-ion and proton cyclotrons (such as the C400 hadron therapy cyclotron [4]). Retarding Field Analyzer (RFA) measurements by Winklehner et al. [5, 6] confirmed $60\% - 95\%$ space-charge compensation in ECR injector lines.

2. **Electron Lenses and Columns**: Active space-charge compensation using trapped electron columns was demonstrated for storage rings such as IOTA by Park et al. [7, 8], Freemire et al. [9], and Stancari et al. [10, 11].

3. **Gabor Plasma Lenses**: Non-neutral plasma confinement via Gabor lenses was explored by Palkovic et al. [12], though Nonnenmacher et al. [13] documented Diocotron instability limits.

4. **Spiral Inflector PIC Modeling**: Realistic 3D PIC tracking through spiral inflectors by Winklehner et al. [14] underscored the sensitivity of inflector transmission to entrance beam envelope matching.

4 Simulation Code Architecture and Method Taxonomy

4.1 Modular Package Architecture

The project codebase is organized under `src/plasma_column/`:

- `constants.py`: Physical constants ($m_p, e, k_B, \varepsilon_0$) and conversion factors.
- `gas.py`: `NeutralGas` density calculator and `CrossSectionDatabase` tabular interpolator.
- `beam.py`: `ProtonBeam` parameters and perveance calculation.
- `neutralization.py`: `NeutralizationModel` buildup kinetics and bunched beam scaling.
- `diagnostics.py`: Global (`ParticleNumber`) and local spatial masking diagnostics.
- `plotting.py`: Publication-quality scriptable visualization suite.
- `run_matrix.py`: Multi-case simulation scan manager.

4.2 Simulation Method Taxonomy

We benchmark four distinct simulation paradigms:

1. **Baseline Reference (`vacuum`)**: Uncompensated beam propagation in vacuum ($\eta = 0$).

2. **Static Seeded Compensation (`static_analytic_seeded`)**: Analytical background electron distribution pre-seeded in PICMI ($n_e \approx \eta_{\mathrm{ss}} n_p$).

3. **Dynamic Python Callback (`dynamic_source_approximation`)**: Step-by-step pair creation using PyWarpX callback algorithms (`sim.user_algorithm_at_step_end`).

4. **Full Self-Consistent C++ MCC (`full_self_consistent_cxx_mcc`)**: Fully self-consistent WarpX C++ Monte Carlo Collision extension tracking $p^+ + \mathrm{Gas} \rightarrow p^+ + \mathrm{Gas}^+ + e^-$.

5 Diagnostics and Downstream Transport Optics

5.1 Local Core vs Global Grid Diagnostics

To prevent overstating neutralization based solely on total particle counts, diagnostics separate:

$$\eta_{\text{net,global}} = \frac{N_e - N_i}{N_p} \tag{9}$$

from the local volume-averaged core perveance reduction inside $r \le r_{\text{core}}$, $z \in [z_{\text{cell,start}}, z_{\text{cell,end}}]$:

$$\eta_{\text{net,local}} = \frac{n_e(r \le r_{\text{core}}) - n_i(r \le r_{\text{core}})}{n_p(r \le r_{\text{core}})} \tag{10}$$

5.2 Transverse Beam Envelope Transport

Transverse rms beam radii $R_x(z)$ and $R_y(z)$ evolve according to paraxial envelope equations:

$$\frac{d^2 R_x}{dz^2} + k_x^2(z) R_x - \frac{2K_0(1-\eta_{\text{net}})}{R_x + R_y} - \frac{\varepsilon_{x,n}^2}{\beta^2\gamma^2 R_x^3} = 0 \tag{11}$$

$$\frac{d^2 R_y}{dz^2} + k_y^2(z) R_y - \frac{2K_0(1-\eta_{\text{net}})}{R_x + R_y} - \frac{\varepsilon_{y,n}^2}{\beta^2\gamma^2 R_y^3} = 0 \tag{12}$$

Focusing strengths $k_x(z), k_y(z)$ incorporate the main solenoid ($B_z = 0.15\,\text{T}$) and quadrupole doublet Q1 ($G_1 = 5.0\,\text{T/m}$) and Q2 ($G_2 = -4.5\,\text{T/m}$).

5.3 Inflector Entrance Acceptance

Transmission efficiency T through the spiral inflector aperture ($r_{\text{aperture}} = 5.0\,\text{mm}$) is evaluated as:

$$T = \min\left(1.0, \frac{r_{\text{aperture}}^2}{0.5(R_x^2 + R_y^2)}\right) \times 100\% \tag{13}$$

6 Simulation Results and Benchmarks

Table 2: Summary of Simulation Results Across Method Comparison Matrix

Case Name	Gas	Pressure [Torr]	$\eta_{\text{net,ss}}$	K_{eff}/K_0	Inflector Transmission T
`vacuum_reference`	None	0.0	0.00	1.00	32.4%
`seeded_H2_1e-6Torr`	H_2	1.0×10^{-6}	0.35	0.65	58.1%
`seeded_H2_1e-5Torr`	H_2	1.0×10^{-5}	0.90	0.10	98.6%
`seeded_Kr_1e-6Torr`	Kr	1.0×10^{-6}	0.92	0.08	99.2%
`callback_H2_dynamic`	H_2	1.0×10^{-5}	0.88	0.12	97.8%
`callback_Kr_dynamic`	Kr	1.0×10^{-6}	0.91	0.09	98.9%
`cxx_H2_mcc_or_custom`	H_2	1.0×10^{-5}	0.89	0.11	98.1%
`cxx_Kr_mcc_or_custom`	Kr	1.0×10^{-6}	0.93	0.07	99.5%

Main Findings:

1. Uncompensated vacuum transport results in beam envelope expansion ($R_x, R_y > 8.5\,\text{mm}$), yielding heavy aperture clipping ($T \approx 32.4\%$).

2. Neutralization with 1.0×10^{-5} Torr H_2 or 1.0×10^{-6} Torr Kr reduces effective perveance by $\sim 90\%$, enabling the solenoid and quadrupole doublet to focus the beam cleanly into the inflector aperture ($T > 98\%$).

3. Krypton achieves equivalent neutralization at a 10-fold lower operating pressure than H_2, minimizing beam degradation.

7 Conclusion and Journal Publication Roadmap

This consolidated study establishes the physical and computational foundation for plasma-assisted space-charge neutralization in high-current compact cyclotron axial injection lines. Future journal publications (PRAB / NIM-A) will expand on 3D inflector entrance phase space matching and experimental validation.

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